

NIKHIL ADAPALA

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461 Brandon Way • Austin, TX, 78733 • (512) 884-7100

EDUCATION

The University of Texas at Austin	B.S. Aerospace Engineering B.S. Mathematics GPA: 4.00	Spring 2029
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EXPERIENCE

The City of Austin I.T. Division • Software Development Intern • Austin, TX	Summer 2025
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- ❖ Architected React front end for TradeRoots, a full-stack skill-exchange platform for Austin community
- ❖ Engineered messaging and reviews using React, Node.js, REST APIs, and SQL for full trade workflow
- ❖ Directed 5-person engineering team through full build cycle, winning 2025 Entrepreneurship Award

HearingTracker • Software Development Intern • Remote	Spring 2024 - Summer 2025
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- ❖ Designed SQL schema managing 500+ hearing aid models across 12 brands with 10+ spec fields each
- ❖ Built React/Node.js app serving personalized recommendations to 30k+ users from 20+ filter criteria
- ❖ Integrated Phonak and Oticon APIs to pull live device specs and pricing into the recommendation engine

PROJECTS

StarForge • Software Project	Summer 2026
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- ❖ Built a procedural 3D space engine, generating full star systems of numerous worlds from one seed
- ❖ Solved floating-point drift on 47,000-unit planets with dynamic origin recentering every 3,000 units
- ❖ Architected multithreaded terrain streaming holding ~100K scatter instances in view at any altitude

Multimodal Deep Learning for Financial Market Prediction • Financial AI Research	Summer 2025
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- ❖ Led a peer-reviewed study benchmarking six architectures for equity price forecasting
- ❖ Cut forecast error 34% and doubled R^2 to 0.68 by fusing market data with FinBERT sentiment signals
- ❖ Designed a custom hybrid architecture and ran ablations isolating top predictive feature contributors

5 D.o.F Robotic Hand • Software Project	Summer 2026
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- ❖ Designed a 3D-printed a robotic hand, modeling all finger assemblies in SolidWorks
- ❖ Programmed an Arduino to independently actuate 5 servos via a custom serial protocol
- ❖ Built two Python pipelines for real-time gesture recognition and speech-driven ASL fingerspelling

N-Body Orbital Mechanics Simulator • Software Project •	Summer 2026
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- ❖ Built a real-time N-body 2,400 line gravitational simulator in vanilla JavaScript and Three.js
- ❖ Implemented a symplectic leapfrog integrator conserving orbital energy indefinitely at 60 FPS
- ❖ Modeled tidal disruption and momentum-conserving mergers with custom GLSL shaders

CERTIFICATES & HONORS

Private Pilot's License (PPL)	Fall 2026 (estimated)
Certified SolidWorks Associate (CSWA)	Summer 2026
Certified Professional Python Programmer (PCPP)	Summer 2025

ADDITIONAL INFORMATION

Programming Languages: Python, Java, C++, JavaScript, C#, R, SQL, HTML/CSS

Libraries & Frameworks: TensorFlow/Keras, Keras Tuner, scikit-learn, pandas, NumPy, MediaPipe, OpenCV, React.js, Node.js, REST APIs, Google APIs, Firebase, PostgreSQL, MySQL, Airtable

Applications & Skills: SolidWorks, MATLAB, Excel, Arduino IDE, Git, Figma, 3D Printing, Soldering

Interests: Piloting, Basketball, Running, Sports, Programming, Mathematics, Learning, Traveling, Geography

Work Eligibility: Eligible to work in U.S. without restriction

LinkedIn: [linkedin.com/in/nikhiladapala](https://www.linkedin.com/in/nikhiladapala)

GitHub: github.com/onedope512